



DEALER BEST PRACTICE SERIES

Off-Board Fluid Filtration

Application Maintenance Site Management Component Rebuild Component Life Management MARC Management

Off-Board Fluid Filtration 1

1.0 Introduction 2

2.0 Best Practice Description..... 2

3.0 Implementation Steps 4

4.0 Benefits..... 5

5.0 Resources Required 5

6.0 Supporting Attachments 5

7.0 Related Best Practices 5

8.0 Acknowledgements..... 6

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

1.0 Introduction

The performance expectations for components and systems on modern mining machines have grown rapidly in the past two decades. This has driven the design of hydraulic systems to operate at much higher pressures, and dramatically increased the load factor on drivetrain components. The increased performance demands also increase the rate of abrasive wear and failure of components compared to older designs, which operated at lower load factors.

2.0 Best Practice Description

2.1 Debris

In order to increase durability, it is necessary to operate components and systems in much cleaner fluids throughout the life of the component. There are three sources of debris:

2.1.1 Assembly Debris

A large amount of debris is often present in machine systems when new machines are assembled. In the past, failure to remove this debris caused a high rate of system malfunctions and component failures at the factory and in early-hour operation. As a result, portable filter carts were designed for use in the factory to clean machine systems before machines were shipped. When used properly, these carts effectively removed assembly debris and cleaned systems to a factory ship target of ISO 18/15. This cleaning of systems prior to shipment dramatically reduced the incidence of early-hour problems.

2.1.2 Break-In Debris

There is a widespread misconception that nearly all system debris is from new machine assembly, and that once systems are cleaned properly, they stay clean. This is not true. Most components produce debris as a result of the normal break-in process. The length of the break-in process and volume of debris produced varies by system. A mining truck provides a good example of how systems vary.

Steering System

Steering systems essentially consist of a small piston pump, steering cylinders and a control valve. Once assembly debris is removed from this system, it produces very little break-in debris and normally stays clean with standard machine filtration.

Transmission

Transmission clutch discs, and gears produce moderate amounts of break-in debris. This process usually takes between 100-200 hours to complete. Even if the transmission were perfectly clean after assembly, break-in debris would still be produced.

Final Drive & Differential

The rear axle produces very large amounts of break-in debris from gears. This process may last up to 8,000 hours. This debris is almost all very small abrasive particles from the hardened gears.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Off-board Fluid Filtration	DATE 22 AUG 06	CHG NO 02	NUMBER 0806-2.10-1002
----------------------------	-------------------	-----------------	--------------------------

2.1.3 Normal Wear Debris

After the break-in process is complete, components still produce microscopic wear particles, but at a much slower rate. The rate is largely affected by how many wear particles are already in the oil. If the oil is contaminated with a high number of abrasive particles, the normal component wear process is greatly accelerated. If the oil is very clean, the normal wear process is much slower, which significantly extends component life.

On-board machine filtration on most mining machines is not capable of maintaining high levels of fluid cleanliness necessary to maximize component life. As a result, portable factory filtration carts are used during PM's and after system repairs. This practice has grown with the increase in MARC contracts and extended component life guarantees.

2.2 Off-Board Machine Filtration (Kidney Loop)

The use of off-board filtration carts started in the factory to remove assembly debris that was causing frequent production line and early-hour problems. The practice eventually migrated to dealers where the carts were initially used to clean systems after a major repair or component replacement. This later evolved into use of carts during PM intervals. Those dealers found that consistent use of the carts for several hours during PM removed large amounts of debris and helped to achieve and maintain a much better level of oil cleanliness.

There are four basic applications of filtration carts in dealer shops and mine sites:

2.2.1 New Machine Assembly:

Large mining machines are too big to ship fully assembled, so large components such as wheel groups are assembled in the field. It is impossible to maintain system cleanliness when major components are open to contamination during assembly.

Even in systems that are shipped assembled (such as hydraulics) cleanliness levels on new machines often exceed the factory roll-off spec of ISO 18/15. This has caused great debate between dealers who claim the system arrived dirty and factory personnel who claim it left clean. There are three reasons for this:

System Not Cleaned Properly

In a few cases, the system cleanliness may exceed the pre-shipment specification, due to inadequate cleaning prior to shipment.

Variability in Particle Counters

Particle counters (whether lab or portable) basically shine a small laser beam through an oil sample and count the number and size of shadows caused by particles in the oil. They also count water droplets, air bubbles and large agglomerated particles oil additives. Inherent variability in both, the particle counter and the sample treatment process, may account for elevated readings of one to two ISO codes. This is problem is even more prevalent in the field use of portable particle counters where cleaning between samples and sample treatment techniques may vary widely. It is not uncommon (due to variability in the particle counter) to deliver varied results (readings) from the same oil sample.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Off-board Fluid Filtration	DATE 22 AUG 06	CHG NO 02	NUMBER 0806-2.10-1002
----------------------------	-------------------	-----------------	--------------------------

Additive Interference

Some new oils contain large amounts of additives. Some of these additive molecules attract and form large clusters in systems where the oil is not be used (such as new systems). These clusters may get large enough appear as a debris particle to a particle counter, and cause the particle count to be higher than it actually is. On new machines, systems should be exercised 1-2 hours to break down these additive clusters before a relatively accurate particle count can be taken.

2.2.2 PM Intervals

Many dealers or customers who are interested in maximizing component life use filtration carts for the major systems during normal PM's. It typically takes 10-15 minutes per system to connect the carts to each system (rear axle, transmission, hydraulics, steering) at the start of the PM. Carts are then allowed to run unattended for several hours while normal PM services are completed.

2.2.3 System Oil Changes:

A long-standing spec for new oil cleanliness has been ISO 16/13. With the increasing level of additives in the oil, filtering oil to this level has become increasingly difficult. In addition, not all sites can justify the cost of permanent recirculating filtration for new oil. A viable alternative is to fill the compartment with new oil and then install and use an off-board filter cart to achieve the desired cleanliness. Machine system contamination is not a problem since the filtration process occurs before the machine is started. It also has the added benefit of removing some system contaminants that would not otherwise be removed if only new clean oil was installed.

2.2.4 Major Repairs

When major components are replaced or systems are opened up for repair, filtration carts should be used to maximize system cleanliness. A leading cause of failure of rebuilt components is failure to clean the system before the new component is put back in service. This is especially true with catastrophic failures, where the system is contaminated with failure debris.

3.0 Implementation Steps

There are several criteria for the number and size of carts required.

Cart Size

Cart filter size and flow rate is determined by the capacity of the system being filtered. As a rule of thumb, the cart should be sized to filter the volume of the system 35 times in a reasonable period of time.

Oil Type

A different cart is required for each oil type. (Example: rear axle oil cannot be mixed with hydraulic oil)

Fleet Size

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Off-board Fluid Filtration	DATE 22 AUG 06	CHG NO 02	NUMBER 0806-2.10-1002
----------------------------	-------------------	-----------------	--------------------------

The number and model of machines being maintained determine the number of carts required. For medium and large fleets, two sets of carts are often required: one set for PM and one for repair. An inadequate number of carts often results in a cart being used for PM or repair and unavailable for use on another machine when needed.

Help with determining the correct number of carts needed, cart size and required tooling is available from the Caterpillar Service Tools Group or from the Marketing and Product Support Contamination Control Group.

Service Tools Group – contact Jim Balfanz, Balfanz_James_W@cat.com

Contamination Control Group – contact Dave Baumann, Baumann_David_L@cat.com

4.0 Benefits

Maintaining oil cleanliness for major components and machine systems increases both reliability and durability. Electro-hydraulic control valves, which are widely used in transmission controls and implement hydraulics, are very intolerant of microscopic, ferrous debris. Heavily loaded wheel and final drive bearings, as well as, duo-cone seals are also easily damaged by abrasive debris commonly found in new oil. Assuring the new oil is cleaned to the desired cleanliness level, and maintaining ISO 18/15 or better significantly reduces the number of contamination induced failures and repairs and significantly extends component life.

5.0 Resources Required

The number and size of filter carts required is determined by fleet size. Carts may be purchased through Caterpillar Service Tool Group, outside suppliers, or built by the dealer. Training for maintenance and operation personnel is also required so that they fully understand the function and importance of the use of filter carts.

6.0 Supporting Attachments

“Improving Component Durability” booklet set- form # SEBF1021.

Consists of one of each of the following:

Fuel Systems	SENR9620
Final Drives and Differentials	SEBF1015
Powershift Transmissions	SEBF1016
Component R&I	SEBF1017
Engines	SEBF1018
Hydraulics	SEBF1019
Managing Fluid Cleanliness	SEBF1020

7.0 Related Best Practices

0808-2.10-1003	-On-board Fluid Filtration
0806-2.10-1000	-Managing Fluid Cleanliness

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Off-board Fluid Filtration	DATE 22 AUG 06	CHG NO 02	NUMBER 0806-2.10-1002
----------------------------	-------------------	-----------------	--------------------------

8.0 Acknowledgements

This *Off-board Fluid Filtration Best Practice* was authored by:
Dick Douglas
Market Consultant
Caterpillar Global Mining
Douglas_Richard_D@cat.com
1-309-675-5699

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Off-board Fluid Filtration	DATE 22 AUG 06	CHG NO 02	NUMBER 0806-2.10-1002